

From Words to Wonder: Designing and Evaluating an AI-Empowered Creative Storytelling System for Elementary Children

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Abstract

While several digital tools exist for children's creative storytelling, few have explored how generative AI can enhance storytelling quality. Our formative research identified five design requirements for AI-powered storytelling tools for elementary students. We developed a system named *StoryPrompt* that enables children to co-create stories and comics with AI, boosting literacy and creativity. Pilot tests with children and HCI experts demonstrated good usability and positive learning experiences. In a mixed-methods evaluation with 40 children from Grades 2-6, we found that *StoryPrompt* significantly improved storytelling creativity and richness, compared to the storyboard method. Observations indicated more purposeful planning and strategic use of AI-generated words and images, facilitating efficient exploration of storytelling alternatives. While children preferred AI images, they recognized the limitations in representing storytelling details. Teacher interviews highlighted the system's motivational potential and classroom flexibility. We discuss the benefits and considerations of using generative AI to enhance creative storytelling for children.

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CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

Keywords

Generative AI, Storytelling, Children, Creativity, Design and Evaluation

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1 Introduction

Creative storytelling is a process where storytellers utilize vocalization, narrative structure, and mental imagery for creative expression and communication [24]. Creative storytelling is a crucial learning activity that can benefit elementary children by enhancing various abilities and skills, such as narration, literacy, social-emotional development, problem-solving, and creativity [12, 13, 24, 34]. Due to its importance, research in human-computer interaction (HCI) has extensively explored the design space of authoring tools to support children's expressive and creative storytelling processes. These tools include interactive applications (apps) and e-books [8, 23], computer games [1], augmented or tangible toys [13, 38], and embodied systems [9], with most designs emphasizing collaborative storytelling practices amongst children. Despite differences in user

interface designs, these tools share a common characteristic of enabling children to participate in the story creation process to a certain extent, by employing various multimedia narrative elements (e.g., characters, settings).

With the advances in artificial intelligence (AI) techniques, recent HCI research has focused on designing AI-based tools to support dynamic, personalized, and effective storytelling [2, 16, 36, 37]. Compared to traditional non-AI interactive storytelling systems, AI tools offer effective ways of generating creative multimodal content based on users' inputs [16, 36]. While several AI storytelling systems for children exist, most have focused on visual thinking or visual storytelling (e.g., generating visual images or creating storytelling scripts) using generative AI tools (e.g., Stable Diffusion, SD) [2, 16, 36]. This visual storytelling approach heavily relies on a child's visual thinking ability. However, not all narrations (e.g., abstract, adjective expressions) and creative imagination can be easily expressed through visuals alone and vice versa. This visual approach may restrict the complexity of narrations, which is important for older elementary children to practice in storytelling activities [29]. More critically, fewer studies have explored if, and how, generative AI could empower children in terms of story quality (particularly in creativity). Therefore, in this study we aim to explore the design and evaluation of AI-empowered interactive systems to support children's high-quality creative storytelling practices.

Based on formative research with 3 elementary teachers and 18 children, we designed and implemented an interactive system named *StoryPrompt* that enables elementary children to co-create creative stories and comics with generative AI, fostering literacy and creativity through storytelling practice. Unlike previous AI visual storytelling tools [2, 16, 36], our system (1) incorporates both *story structure planning* and *emergent storytelling* allowing children to create complex narratives; and (2) uses both text-based and visual generative AI to stimulate creativity through the words, problem-solving, and visual inspiration and engagement (Fig. 1). Pilot testing with eight new children and three HCI experts demonstrated good system usability and satisfactory learning experiences. We further refined the design and evaluated it with 3 elementary teachers and 40 children in Grades 2-6 who did not participate in the formative study. We conducted a within-subjects study where children were randomly assigned to two groups, with one group using the AI system, followed by a traditional paper-based storyboard method, while the other group experienced the reverse. We invited five experts to blind evaluate children's storytelling quality based on the Consensual Assessment Technique [9]. The results showed that children using the AI system achieved significantly higher storytelling quality, particularly in creativity and richness. Qualitative observations and interviews revealed more purposeful storytelling planning and strategic use of AI-generated words and images, allowing children to efficiently explore storytelling alternatives by ideating, expanding, or modifying the content. Children in both conditions expressed preferences for visual images but also noted their limitations in storytelling (e.g., inefficiency in drawing, inaccuracy by AI), suggesting the inefficiency of relying on visual storytelling as a dominant approach. Interviews with three teachers highlighted the AI system's potential to motivate learning and its flexibility for classroom integration. We discuss the

design considerations and advantages of leveraging both text-based and visual generative AI to support children's effective creative storytelling.

This study contributes to the child-centric AI design community in three ways. First, the design rationale and implications derived from the findings can benefit future research in creating AI-empowered creative storytelling systems for children. Second, we present a research prototype that leverages both text-based and visual generative AI to support children's storytelling for complex narration. This demonstrates the flexibility of AI-empowered systems in supporting children's creative education across grades. Third, the evaluations demonstrated the effectiveness of creative storytelling quality with the AI system compared to the storyboard method. This revealed various similarities and differences in children's emergent storytelling strategies within the two methods, highlighting their diverse planning and strategic use of AI texts and visuals in storytelling. The results suggest the importance of offering both text-based and visual generative AI resources in storytelling and balancing planning and emergent storytelling while providing balanced autonomy for children and AI during their collaboration.

2 Background

We reviewed literature on children's storytelling practices to understand the critical elements and methods to incorporate into the design. We also analyzed the design of interactive storytelling systems, particularly AI systems, to identify design opportunities and evaluation methods for children's creativity as study design references.

2.1 Children's Storytelling Practices

The research has highlighted that children's capabilities and preferences in storytelling practices, reveals that children as young as two can begin to understand *story structures*, such as using “*Once upon a time*” as a formal opening, and “*They all lived happily ever after*” as a formal close [32]. By age four, children can develop a *sense of story*, including *characters*, *settings*, and *plots* [14]. While characters, setting, and plots are considered central elements in storytelling, other researchers have also argued for the inclusion of *emotions* [25, 29], *props* [16, 18], and *actions* [16, 18]. The *character* is suggested to be the most significant, as character traits and actions can influence the flow of story plots and other aspects [4], while different visual representations of the characters (avatars) can impact the story quality and creativity [21, 34]. It seems that children have an innate love for fairy tales and prefer to use their personal experiences as storytelling materials [34]. The inclusion of nonsensical and fantasy-based narrative elements not only engages children but also inspires their creativity and imagination [1].

Various non-digital tools, activities, and strategies are often used in instructions to support children's effective storytelling [6, 35, 37]. Physical toys, visual pictures, drawing activities, and embodied enactment (e.g., fantasy play) are suggested to be effective in engaging children in visual thinking and embodied storytelling experiences [6]. Another frequently used strategy is *guided conversation*, where a storyteller asks children questions about story content and provides responsive feedback [35, 37]. This strategy highlights the

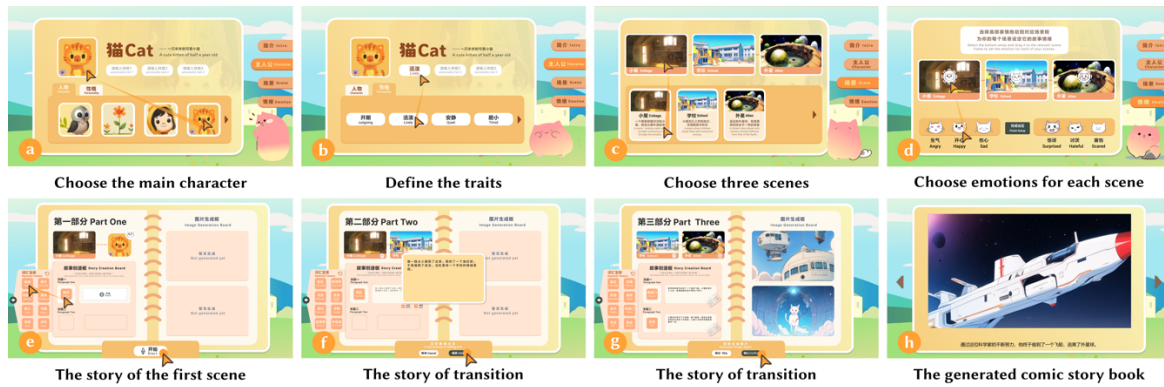


Figure 1: The user interfaces (UIs) of *StoryPrompt* that supports elementary children’s creative storytelling with generative AI. The high-resolution file can be accessed at https://drive.google.com/file/d/1lwOCqmQbuaNCdXcysBg_2EAAUHVW496s/view

importance of involving in reading experts (e.g., teachers, system-guided instruction) to facilitate collaborative storytelling among children. Many studies have demonstrated the effectiveness of storytelling practices in enhancing a wide range of skills in children. These enhanced skills can be broadly divided into three categories: *creativity* (e.g., problem-solving skills, imagination, creative storytelling) [1, 34, 36], *literacy* (e.g., narrative knowledge, languages, writing skills) [22, 32, 37], and *social-emotional development* (e.g., empathy) [34, 37]. There are other associated benefits including increased learning interest, motivation, and memorization [28, 36, 37]. In this study, we focus on literacy (story quality) and creativity (story creativity) as these are particularly important for elementary children in various grade levels, and it is believed that social-emotional development is typically acquired during preschool or early elementary stages.

2.2 Design of Interactive and AI-based Storytelling Systems for Children

Many digital storytelling systems for children have been designed based on different interactive storytelling models: *emergent storytelling*¹ [23, 27, 31], *character-driven storytelling*² [4, 5, 21], and *user-centered plot resolution*³ [7, 33, 38]. For example, *Graphic Storywriter* is an interactive computer software that supports children’s emergent storytelling [32]. This software allows the child to select story scenes, and to drag characters and objects onto the story stage, after which, the associated texts are generated. *Story Creation* is a character-driven storytelling system that allows children to use an avatar to enact a story [34]. The *Interactive Storytelling Robot* is a tangible system that allows children to make decisions on several narrative branches of a predefined story through question-and-answer dialogues with a robot [19].

Recent research has explored the use of AI techniques to support children’s collaborative storytelling, with a particular focus on visual storytelling. Bernal et al. designed *Paper Dreams*, a web-based

platform that uses AI to convert users’ text into images [2]. They argued that the real-time generation of visuals based on multimodal input can encourage divergent thinking and story development while also ensure a user’s ownership. Han and Cai presented *AI Story*, a web-based app that allows children to create a simple visual story with AI [16]. They conducted formative research and proposed design considerations such as safety and ownership. The app allows the child to select predefined stickers (characters, facial expressions, backgrounds, props, and effects) and records the image caption, after which the AI art generator enriches the details and generates the final image. Zhang et al. presented *StoryDrawer*, a visual storytelling system that uses AI to transform children’s real-time storytelling into drawings and generate abstract sketches with semantic similarity to existing content [36]. The results showed that *StoryDrawer* provoked children’s creative and elaborate ideas, contributing to their creative outcomes during an engaging visual storytelling experience. However, the three designs heavily rely on a child’s visual thinking ability (drawing all the elements in a single page), which may limit complex sequential storytelling. Another system presented by Zhang et al., called *StoryBuddy*, enables parents to collaborate with AI in creating question-and-answer sessions for children [37]. This project aims to use AI to generate questions and facilitates guided conversation during parent-child storytelling processes. Liu et al. proposed five-week story making sessions with seven families and a therapist who used generative AI and traditional materials for expressive arts therapy. The results demonstrated the therapeutic meanings of generative AI and suggested the importance of materializing it in family-AI engagement [20].

2.3 Evaluation of Children’s Creativity and Creative Storytelling with Systems

Creative storytelling is a common method to understand a child’s creativity. Researchers have proposed various ways to conceptualize and evaluate creativity. Boden distinguished between historical creativity (ideas never before conceived) and psychological creativity (ideas new to an individual) [3]. Rhodes introduced the “four Ps of creativity” model [26], highlighting four levels of creativity: *Person* (personality traits), *Product* (characteristics of outcomes),

¹The user creates a story by selecting certain narrative elements (characters and backgrounds) from a large corpus of materials such as [31].

²The user plays as one of the characters and enacts the story [4].

³The user participates in the storyline and influences dynamically the development of the plot through decision making process [30].

Process (generating and selecting alternatives), and *Press* (environmental impacts). Based on these four levels, three main approaches are commonly used to evaluate creativity, including *physiological measurement* (e.g., Alternative Uses Task), *self-reporting* (e.g., Creative Achievement Questionnaire), and *expert judges* (e.g., Consensual Assessment Technique, CAT) [15]. Physiological measurement and self-reporting often assess the ‘Person’ and ‘Process’ levels, while expert judges evaluate ‘Product’ and ‘Press’. Researchers also highlighted the interplay between individual and environmental factors in creativity. Kupers et al. emphasized the role of emergence and constraints, advocating a shift from macro-static to micro-level measuring processes to capture children’s dynamic creative strategies with environments (like technologies) [17].

In children’s creative storytelling with interactive systems, many HCI studies have focused on both the ‘Product’ level (storytelling quality) [9, 10], and the ‘Process’ and ‘Press’ levels (interaction with systems) [15]. Chu et al. examined 20 children’s creative self-efficacy and thinking through storytelling by assigning them to use either an embodied storytelling system or a comic drawing method. They employed the CAT with five experts rating children’s storytelling quality from the dimensions of creativity, fluency, richness, and story structure [9]. Elgarf et al. measured children’s storytelling fluency, flexibility, elaboration and originality by evaluating story elements and categories [10]. Beyond storytelling quality, researchers explored children’s storytelling processes with systems to understand intuitiveness, usefulness, feasibility, and engagement of interactive systems in supporting creative storytelling [15].

In summary, despite the focus on digital storytelling in HCI, there is limited exploration of generative AI tools for co-creating stories with children. This leads to our core research question (RQ): **How can we design and evaluate a generative AI storytelling system that effectively supports co-creation with children, enhancing literacy and fostering creativity?** This inquiry aims to bridge the gap by leveraging AI to enrich the storytelling experience.

3 Formative Research

We performed formative research involving 3 elementary teachers and 18 children to understand current storytelling educational practices and derive design implications. The studies were approved by our university’s ethical review board (same as the following studies). Specifically, one of the co-authors who works in the elementary education system helped contact the voluntary local school. Study information flyers were sent out by the school to the teachers, children’s guardians, and children. Participation was based on voluntary informed consent from the teachers and children’s guardians, and the assents of the children. During the assent process, researchers informed the children of the study tasks, explained that they would be audio recorded and photographed, and assured the children that they could stop at any time. If the children verbally assented, the researchers started the study. All data was anonymized and stored in secure locations.

3.1 Methods

We conducted in-depth interviews with 3 female teachers who had 10 years of teaching experience in the subject of Chinese. The interview aimed to understand current storytelling instruction in

schools, including learning goals, effective training techniques, and potential challenges. Each interview lasted around 30–60 minutes and took place in a quiet classroom. Two researchers participated, with one primarily responsible for asking questions and the other for taking notes based on our study protocol (Appendix A. 1). The interviews were audio recorded, and each participant received \$30 as research compensation. We also conducted 3 rounds of focus groups with 18 children, involving 9 boys and 9 girls from Grade 2 to Grade 4, with an average age of 8.00 ± 0.84 . The focus groups aimed to understand children’s prior experiences with computers and AI, their storytelling strategies, and their preferences for story characters and settings (Appendix A. 2). Each focus group contained six children and two researchers, with one researcher serving as the moderator (asking questions) and the other as the facilitator (taking notes). Each round lasted 30–40 minutes, taking place in a quiet classroom, with each focus group being recorded and photographed. Each child received a ‘Maker suite’ as a gift after the study. The audio recordings of the focus groups were transcribed into text, and then the three researchers used a thematic approach to analyze the qualitative audio data by identifying similar themes.

3.2 Key Findings

All the teachers agreed that creative storytelling is crucial for children, especially in developing essential creative writing skills. Several aspects were highlighted: **(1) Learning goals:** younger children (Grades 1–3) focus on learning literacy words, story retelling techniques, and telling simple stories from personal experience. In contrast, older children (Grades 3–6) concentrate more on developing the ability to create stories with complex structures. **(2) Effective training techniques:** guided instructions are crucial to support children’s effective learning. This could involve asking *purposeful questions* (e.g., “If you were the character, what would you do?”), *offering literacy prompts* (e.g., hints for words or phrases), *providing visual aids* (e.g., concrete pictures to describe), and *relating to other references and materials* (e.g., books the children had read before). Having a general plan for the story structure while allowing for emergent storytelling and iterative fine-tuning is important for creating a high-quality story. Multimedia materials, such as images, can stimulate children’s learning interests. **(3) Potential challenges:** beyond reading, there were limited materials to inspire children’s creative storytelling and writing. Finding suitable digital storytelling tools for various grades was challenging. **(4) Other feedback:** elementary schools have digital equipment, such as computers and iPads, and children of that age are familiar with computer functions.

Regarding a child’s preferred characters and settings, we identified four main character types and four main settings. The four types of characters were: *talking animals or objects* (e.g. talking cats), *people with magic or superpowers* (e.g., witches), *sci-fi people* (e.g., scientists), and *normal girls or boys*. The four types of scenes included: *natural scenes* (e.g., forests), *fantasy scenes* (e.g., castles), *sci-fi scenes* (e.g., labs), and *everyday scenes* (e.g., schools). We also noticed that higher-grade children preferred to choose sci-fi and fantasy-related scenes and characters, while lower-grade children liked objects, or animals, or themselves. In terms of gender differences, boys showed more preference toward sci-fi scenes and

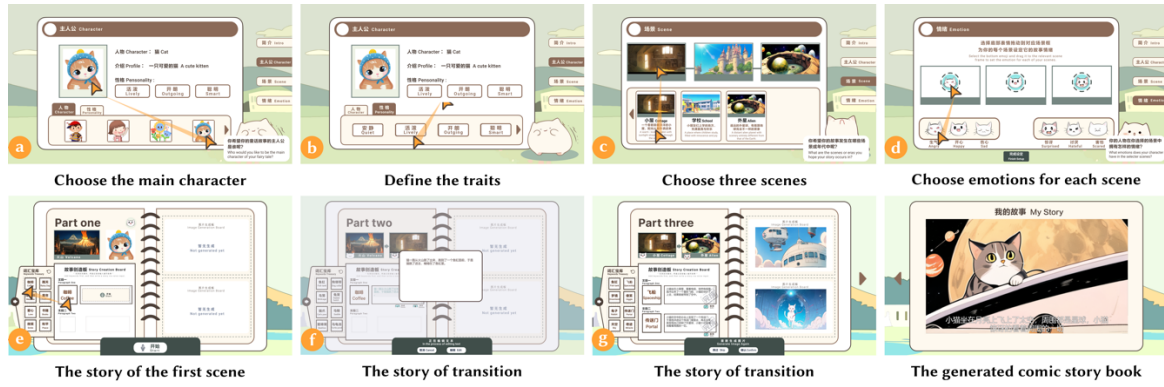


Figure 2: (a) Selecting the main character by dragging the image into the profile box; (b) defining the character personality by dragging three labels into the boxes; (c) selecting three scenes by dragging the images into the boxes in sequence; (d) defining the three associated emotions for each scene by dragging the emotion icons into the boxes; (e) selecting the AI-generated keywords by dragging them into the boxes and using the keywords to create storylines for the 1st scene; (f) using voice recording to generate the storyline and modifying it using text input for the 2nd scene; (g) generating AI images for the 3rd scene; (h) viewing the entire generated storybook. The high-resolution file can be accessed at https://drive.google.com/file/d/1fCeyeytA0gF744Cw1-04MJfRUwVV22ys/view?usp=drive_link

characters compared to girls. These findings helped us to identify five design implications and provided useful examples (characters, settings) for the design.

3.3 Design Implications

Based on the empirical results, we have identified five design implications (DI). **DI#1:** the system should support varying difficulty levels of creative storytelling activities for children across different grades. **DI#2:** the system should allow children to plan the story structures while also opening possibilities for emergent storytelling, serendipitous discovery, and iterative creation. **DI#3:** generative AI could be used to provide both verbal (words) and nonverbal (visual images) prompts to inspire children’s creative storytelling and learning interests. **DI#4:** purposeful guidance is important in children’s storytelling. **DI#5:** story characters and settings should be designed based on story genres and children’s preferences.

4 The AI Storytelling System Design

Based on the design requirements, we designed a system called *StoryPrompt* that enables elementary children (ages 7–12) to co-create creative stories and comics with generative AI, and we will now describe the interaction flow, main features, and technical solution.

4.1 Prototype Overview

The system runs on a computer and can be used by an individual child or by a group of 2–3 children. While interacting with it, the child begins by selecting a character (Fig. 2a) and defining the character’s personalities (Fig. 2b). Next, the child selects three sequential story scenes (Fig. 2c) and defines three character-emotions associated with each scene (Fig. 2d). Once the story setup is complete, the main task is to fill the story gap (make a reasonable and coherent story) by relating the AI generated keywords with the

other story elements (Fig. 2e–f). Specifically, the child needs to select inspiring keywords (nouns) from a set of words generated by ChatGPT to create two storylines (two paragraphs) for each scene (Fig. 2e). The storylines could describe how the character and the objects (keywords) come to the scene and why the character experiences such emotions. The child can record their voice to generate the text and then modify the details using text input (Fig. 2f). The generative keywords are then updated in real-time based on the story setup and the children’s story input. After each storyline is created, the system generates a comic-styled image using our trained SD model (Fig. 2g). The child needs to complete a total of six storylines for three sequential scenes to finish the entire story. At the end of the activity, the child can view and download the generated comic storybook consisting of six pages (Fig. 2h). From findings, we recommend that children below Grade 3 use the system with a teacher.

4.2 Design Features

4.2.1 Story Characters, Settings, and Emotions. The child first plans the story structure by selecting one main character, three settings, and three associated emotions (DI#2, Fig. 2a–d). We incorporated four types of ten characters and four types of seven settings based on our formative study (DI#5). Various settings and emotions are used to trigger the story transition because changes of character locations or emotions often predict the story development and can serve as a good starting point for children to expand their imagination. The system is flexible in supporting the selection of same or different settings or emotions, enabling simple or complex story generation (DI#1).

4.2.2 AI Generative Keywords as Narrative Prompts. In each setting, the children can click the left sidebar to view the eight keywords generated from ChatGPT (DI#3, Fig. 2e). A child’s main task is to drag the selected keyword into the squared box and create the

storyline by relating this keyword to the story elements (ID#2). If the child is unsatisfied with the keywords, they can update the keywords by clicking the icon. The initial eight keywords are generated based on the selected character, settings, and emotions, while the updated ones are generated in real-time from the child's input. Among the words, four are strongly associated to the storyline (e.g., generating the words “queen” and “crown” when a child says, “a princess is reading in the garden”), while the other four are not (e.g., providing the words “moonlight” and “treasure” when a child says “a cat walks in the forest”). This design accommodates the varying difficulty levels of the storytelling tasks (ID#1).

4.2.3 AI Generative Visual Images. The child creates the entire story by completing six story paragraphs (storylines) one by one (2 paragraphs per setting, $\times 3$ settings) (Fig. 2). Once the child is satisfied with each storyline, they can click the generation button to create its image (DI#3) (Fig. 2g-h), and the child can regenerate the image until satisfied.

4.2.4 Others. The system incorporates a set of questions based on teachers' interviews to provoke meaningful dialogues (ID#4). Several questions include “how did the character arrive at this scene? Why does the character have such an emotion?”. Teachers can read the questions aloud to younger children while older children can use them for self-guidance. Story input can be done through voice recording or text input, making it accessible for children of different ages (DI#1).

4.3 Technical Implementation

The system was implemented using Unity. For real-time AI keyword generation, we employed prompt engineering and provided cases to control ChatGPT. This enabled us to generate four strongly associated keywords and four weakly associated keywords that were relevant to users' input text and appropriate (non-violent and free of mature themes) for elementary-aged children. For real-time AI image generation, our process involved calling the ChatGPT API to extract key characters, scenes, and actions from the corresponding text while adding details. We then utilized the SD API to generate images based on the extracted information. To achieve aesthetic, cartoon-style images appealing to children, we trained our SD model. Theme keywords, lens, style references, adjectives, and suffix words were used to create a formula inspired by Midjourney to generate the SD training materials and maintain visual style consistency. Finally, voice recognition and audio-to-text translation were implemented using the Baidu API.

5 Preliminary Evaluation and Design Iteration

Before examining the learning effectiveness, we conducted a preliminary study with eight children and three HCI experts to assess the system's usability and users' learning experiences. The participant recruitment and consent process were like the formative study, but we specified that children would use an AI tool developed by our lab, individually or collaboratively, for creative storytelling facilitated by the researchers, and this process would be videotaped.

5.1 Preliminary Evaluation

Eight children (4 boys and 4 girls, Grades 1-5, average age 8.63 ± 2.13) participated in the pilot testing. All had experience with apps, but only three had used AI products, with four children using the system alone, and four using the system in pairs. Primarily, a researcher introduced the study and demonstrated the system, then children created stories and rated the system design, story quality, and generated images on a scale of 1 to 10. Participants also provided feedback on likes and dislikes in the post-interviews. The study was conducted in a quiet classroom, lasting 20-40 minutes per session. Three researchers facilitated this research study, set up equipment (the AI system and video cameras), and took observational notes. There were also HCI experts who specialize in child-centric and AI design, who reviewed the system providing feedback on design improvements (Fig. 3). As a preliminary usability test, the main purpose was to identify any potential design or technical issues, and to ensure that the system could be used as a research instrument for further study. Therefore, we did not conduct rigorous measures, but instead, we synthesized qualitative observational and interview data, coding based on three themes: learning curves, use experiences (i.e., attention and engagement), and any challenges that arose. We also summarized the key feedback provided by the HCI experts.

The primary results include the following. (1) All children quickly learnt the system, although they needed more explanation about the emotion UI page to clarify the relationship between the character emotion and scene. (2) Children stayed focused and seemed engaged during the process. (3) The system supported individual and pairs of children to create six stories with a mix of narrative genres, including a total of 1830 words and 36 images. However, younger children demonstrated challenges in managing three different scenes and three different emotions. Older children created longer and complex story texts which overlapped with the image. All the children felt challenges in integrating the AI keywords initially but became more confident as the process proceeded. (4) Children perceived this storytelling method (8.75 out of 10) as higher quality and their co-created story with AI (8.75), opposed to the generated AI images (7.75). Yet, children preferred the functions of AI-generated images and keywords. (5) The HCI experts acknowledged the design as beneficial for creative storytelling but suggested adding audio instructions for navigation and sound effects to enhance engagement. The detailed results can be found in [11].

5.2 Design Iteration

Based on the results, we further improved the system by: (1) re-designing the UI pages to better illustrate the relationship between emotions and scenes; (2) improving navigation and adding sounds effects; (3) scaling down the text font size and separating the generated story text from the image. We also revised the ‘system use protocol’ by recommending that children view the entire interaction flow before starting to tell the story and begin with 1-2 story scenes and emotions (Fig. 4).



Figure 3: (a) The system used by an individual child (Jessica); (b) the system used by children in pairs (Lisa and Robert); (c) a part of the storybook created by an individual child (Jessica); (d) the storybook created by children in pairs (Lisa and Robert).

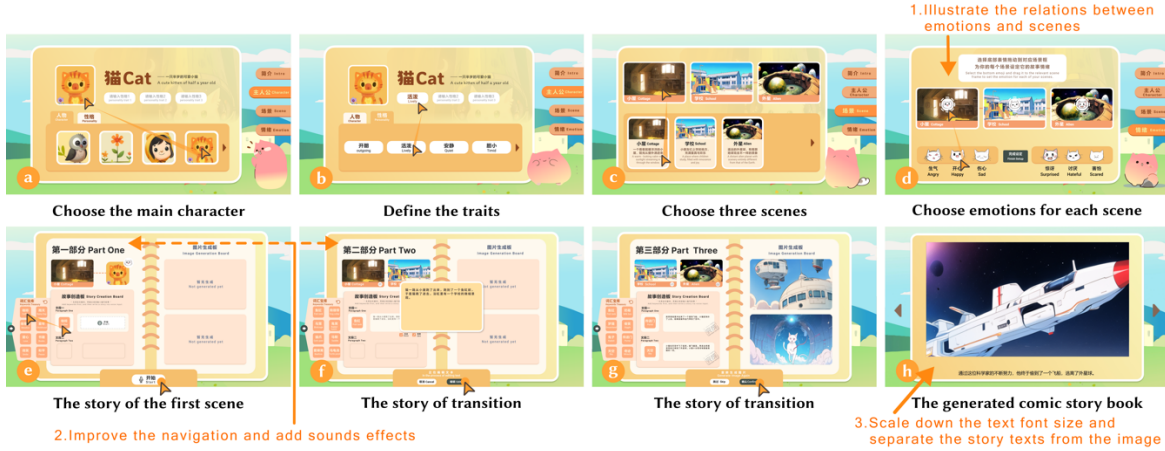


Figure 4: The redesign of the system UI.

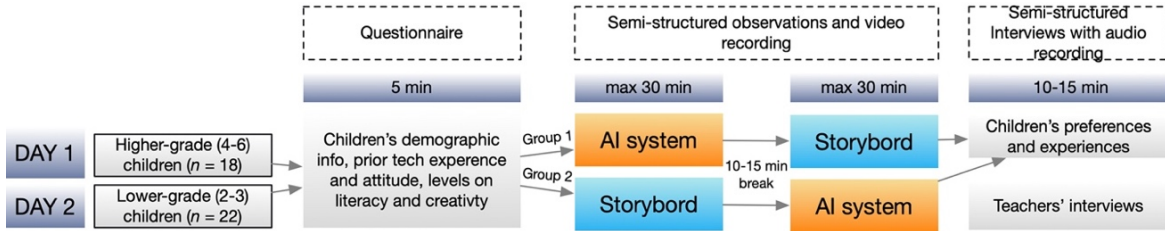


Figure 5: The within-subjects study design.

6 System Evaluation

To investigate if and how *StoryPrompt* supports children’s creative storytelling, we conducted a within-subjects experiment with 40 children using both the AI system and traditional paper-based storyboard and drawing methods (Fig. 5). We chose this study design for two reasons: (1) the paper-based storyboard and drawing is commonly used in preliminary schools and has served as a control group in previous studies [9, 32]; and (2) a with-subjects design helps to control personal characteristics such as age, gender, creativity, literacy, and learning styles, as well as ensuring that all participants can collaborate with AI, hence promoting educational equity and ethics. To control order and overlearning effects, children were randomly assigned to two groups: one used the AI system first (AI condition), then the storyboard method (Storyboard condition) and the other did the reverse. Our three research questions are:

(Rq1) What is the level of creative storytelling quality in both conditions, and do time and order affect outcomes (*creativity product outcomes*)? (Rq2) What are the creative storytelling strategies in both conditions (*creation process strategies*)? (Rq3) What are the children’s preferences and teachers’ attitudes towards the methods (*preferences and attitudes*)? The consent process was similar to the preliminary study, except that each child would use the AI system individually.

6.1 Participants

We attempted to include primary school students from various grades, and with diverse prior AI/technological experiences as much as possible. Forty children from a public elementary school voluntarily participated in the study, including 30 boys and 10 girls, with an average age of 9.64 ± 1.18 years (Grade 2-Grade 6). All had prior experience with computers or iPad apps, and most ($n = 31$)



Figure 6: (a) The lecture on creative storytelling; (b) the AI condition; (c) the storyboard condition.

had explored AI products like AI drawing, writing, or voice-based tools. The children rated their creativity: 32.5% excellent, 42.5% good, 22% average, and 2.5% weak. For storytelling ability, 25% rated themselves excellent, 40% good, 25% average, and 10% weak. Their general attitude towards AI products averaged 4.50 ± 0.64 on a 1-5 scale indicating positive views. Three elementary teachers (two females aged 38 and 26, and one male aged 48) with 13, 1, and 38 years of experience, respectively, observed the study and were interviewed.

6.2 Procedure

The study was conducted over two days in quiet classrooms, and with higher-grade children (Grades 4-6) participating on the first day, and lower-grade children (Grades 2-3) on the second day. The studies started with a 30-minute introduction to creative storytelling by a trained teacher (Fig. 6a). The children were then randomly divided into two groups: with one using an AI system (Fig. 6b), while the other employed the traditional storyboard (Fig. 6c). The children were asked to select one main character from the eight options given (i.e., girl/boy, robot, flower, scientist, cat, witch, and princess), and to choose no more than three scenes from seven available options (i.e., lab, castle, volcano, school, forest, cabin, and outer planet) to complete a creative storytelling task using the designated method for their condition. Each child had 20-30 minutes for their task followed by a break, then they switched methods. We asked the children to think about a new story the second time but did not enforce it if they could not. Eight trained facilitators supported the process: four for the AI system, two for traditional storytelling, and two for managing children and taking photos. All researchers were responsible for observations and interviews. All facilitators had a study protocol with scripts to ensure consistency.

Specifically, when using AI, children were first given a system tutorial to understand how to use it, then they used the system for storytelling mainly via voice recording and text typing, with a facilitator nearby to assist with any technical problems. The process generally followed the system interaction as depicted in Fig. 2 and Fig. 4. After completion, the facilitator helped review the entire story and the six generated images, making any necessary changes required. When using the storyboard, children first received paper-based storyboards with six slots for drawings and text, markers, pencils, and printouts of characters and scenes. They were asked to select characters and scenes, sketch ideas, and write notes for storytelling with facilitators aiding as needed. Once the children felt that they had completed their stories, they presented them

orally to facilitators, who recorded and converted them into text helping to correct any errors.

After each condition, children were interviewed about their storytelling strategies, use experience, and preferences for the two methods, with some students voluntarily sharing their stories with the class. Children's demographic information was collected, and each child received a toolkit as a gift. The three teachers observing the study shared their perspectives on the two methods.

6.3 Data Collection and Analysis

We used a mixed methods approach for data collection and analysis. Children's storytelling artifacts, including the story texts (converted from recorded audios) and generated visuals (children's drawings and the images created with AI, Fig. 7), were collected, scanned, and stored anonymously. To further evaluate story quality, we used the CAT method by inviting six experts to rate children's storytelling quality individually. The experts included one elementary school teacher in Chinese literacy, two majors in creative writing, and three in education and HCI. The experts rated the stories (mainly texts) on four dimensions based on the children's ages: *creativity* (how original and imaginative the story is), *richness* (the depth and detail of the story), *fluency* (flow and ease of reading), and *narrative structure* (how well the story is organized with a logical narrative development). A scale from 1-5 was used, with 1 as the lowest quality and 5 as the highest [9], without knowing which creation method was used (blind review). They also rated the discrepancy between the stories created by the AI system or the storyboard for a same child (on a scale from 1-5, with 1 as the smallest discrepancy and 5 as the largest). The experts first briefly discussed the codes and individually coded 18% of (14 out of 78 stories) of the materials, achieving a level of 92.86% consistency. They then held a one-hour online meeting to resolve inconsistencies and establish rating standards, and after this, five experts⁴ independently rated the scores.

A questionnaire was used to collect children's demographic information and personal characteristics (e.g., their perceived levels of creativity and storytelling). The children's interaction process with the AI system was videotaped via screen recording, while their traditional drawing and storytelling behaviors were photographed. Semi-structured observational sheets were used to collect children's behaviors and strategies, and semi-structured interviews helped to understand their reasons, experiences, and preferences (Appendix

⁴One expert who specializes in creative writing did not participate in the final coding.



Figure 7: Examples of children's storytelling visuals using storyboards (a-P28, b-P40) and the AI system (c-P33, d-P30).

B. 1). The teachers were also interviewed to acquire their attitudes towards the two methods (Appendix B. 2).

For Rq1, we first analyzed the story genres and calculated the wordcounts in the AI and Storyboard conditions, using paired-samples *t*-tests to compare the word differences between the two. Additionally, we converted all five experts' ordinal scores to continuous scores and calculated the average mean for each dimension and for all the four dimensions (i.e., referred to as "storytelling quality"). We then used quantitative methods to: (1) compare the average mean of children's storytelling quality and the four sub-dimensions between the AI and the Storyboard conditions with paired-samples *t*-tests, and (2) compare *time* (time 1: AI & Storyboard, or time 2: AI & Storyboard), and (3) *order* (AI first or Storyboard first, Time $\times$ Condition) effects on storytelling quality by using paired-samples *t*-tests and two-way ANOVA. **For Rq2**, we mainly used thematic analysis to code the observations, video data, and interviews to understand children's behavioral patterns and their underlying reasons. This was supplemented with descriptive analysis (percentages) of children's strategies on selected characters and scenes. **For Rq3**, we used thematic analysis for analyzing children's preferences and teachers' feedback in the interviews; three researchers coded the qualitative data.

7 Results

In this section, we present the results of our analysis of children's storytelling quality, their strategies during the process, and their preferences and experiences with the two methods, and we also briefly report on teachers' attitudes.

7.1 Children's Storytelling Quality

7.1.1 Storytelling Genres and Words. The children in the storyboard condition successfully created 38 complete stories⁵ (6840 Chinese words in total), including eight types of genres such as *fantasy*, *sci-fi & adventure*, and *fantasy & adventure*. Children in the AI condition created 40 complete stories (10075 words in total), with seven types of genres such as *fantasy & adventure*, *fantasy*, *sci-fi & adventure*. An additional genre of *realistic fiction* was found in the storyboard condition, with three children presenting their stories in a diary style. A paired-samples *t*-test revealed that the word count of stories created with the AI system was significantly higher (249.132 ± 100.698) than those from the storyboard method

⁵Two children (P36 and P39) did not complete their stories in the storyboard condition. The absence of their data violated the normal distribution, so we removed their data from later statistical analysis. Additionally, since these two participants did not follow our pace, we missed collecting their interview data after the traditional condition and did not include them in the quantitative analysis later. However, their qualitative data were included in the AI condition.

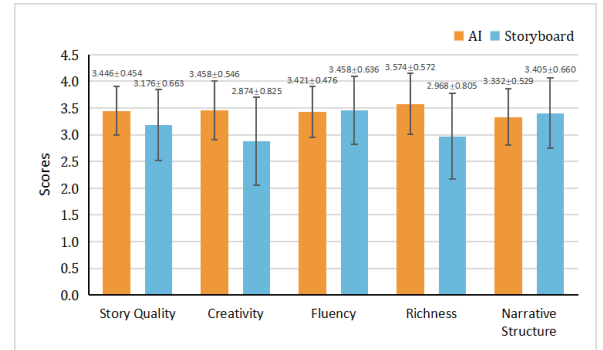


Figure 8: Children's storytelling quality scores in the AI condition and the storyboard condition.

(180.00 ± 86.332), with a mean increase of 69.132 words (95% CI, 35.407 to 102.856), $t(37) = 4.153$, $p < .001$, $d = 0.67$.

7.1.2 Storytelling Quality, Creativity, Richness, Fluency and Narrative Structure. A paired-samples *t*-test showed that children in the AI condition achieved significantly higher story quality scores, particularly in creativity and richness, compared to the storyboard condition. The AI stories had a higher mean score of 3.446 ± 0.454 , compared to 3.176 ± 0.663 for the storyboard, showing a significant difference of 0.270 (95% CI, 0.053 to 0.486), $t(37) = 2.524$, $p < .05$, $d = 0.41$. Creativity scores were higher in the AI condition (3.458 ± 0.546) other than the storyboard (2.874 ± 0.825), with a mean increase of 0.584 (95% CI, 0.288 to 0.880), $t(37) = 3.997$, $p < .001$, $d = 0.65$. Richness scores were also higher with AI (3.574 ± 0.572) versus storyboard (2.968 ± 0.805), with mean increase of 0.605 (95% CI, 0.323 to 0.887), $t(37) = 4.349$, $p < .001$, $d = 0.71$. When considering fluency (AI: 3.421 ± 0.476 , Storyboard: 3.458 ± 0.636) and narrative structure (AI: 3.332 ± 0.529 , Storyboard: 3.405 ± 0.660), the storyboard condition outperformed AI, but differences were not significant (Fig. 8).

We also evaluated impacts of *time* on storytelling outcomes using paired-samples *t*-tests. No significant results were found ($t(38) = 0.491$, $p = 0.627$, $d = 0.796$), although the second phase (3.283 ± 0.566) was slightly lower than the first (3.340 ± 0.600), possibly due to children's fatigue and their decreased interest in performing similar tasks. However, the AI system always outperformed the storyboard method in storytelling quality, both in the first (AI: 3.432 ± 0.483 , Storyboard: 3.264 ± 0.683) and the second (AI: 3.457 ± 0.441 , Storyboard: 3.068 ± 0.641) phases, suggesting its effectiveness in supporting high-quality storytelling.

We further conducted a two-way ANOVA to examine if *order* (AI first or Storyboard first) influenced storytelling quality. Results showed no significant ‘Time (phases1/2) × Condition (AI/Storyboard)’ interaction on story quality, $F(1, 72) = 0.425$, $p = 0.517$, $\eta^2 = 0.006$. Contrary to our expectations that children who experienced AI first might transfer learned techniques into the storyboard condition, children starting with the storyboard method scored slightly higher storytelling quality (except for fluency) in both AI and storyboard conditions. Based on our observations, we infer this might be due to children’s increased task familiarity and motivation with the AI system following the storyboard method. Conversely, children starting with the AI condition showed decreased interest in the storyboard method, leading to lower performance. This was consistent with findings that: (1) children using the storyboard method second had the lowest scores (3.068 ± 0.641) compared to the other three conditions (AI first, AI second, and Storyboard first); and (2) children first experiencing AI followed by the storyboard method scored lower on story discrepancy (3.188 ± 0.121) than the reverse order (3.629 ± 0.991). This indicates that there are less effort and engagements in storyboard storytelling after using AI.

7.2 Children’s Storytelling Strategies

7.2.1 Storytelling Strategies with Storyboards. In the storyboard condition, the top three selected characters were a *little boy* (28.95%, $n = 11$ out of 38), a *scientist* (18.42%, $n = 7$), and a *little girl* (15.79%, $n = 6$), while top three selected scenes were the *forest* (23.68%, $n = 27$ out of 114), *cabin* (19.30%, $n = 22$), *lab* (8.77%, $n = 10$) and *castle* (8.77%, $n = 10$). Half of the children actively or partially planned the story structure in advance ($n = 20$). Several children ($n = 6$) began planning when they started to draw, while the rest ($n = 12$) did not plan at all. A total of 63.16% of the children ($n = 24$) indicated that they treated themselves as the characters and used personal experiences as storytelling references, with some mentioning inspirations from animations, novels, movies, and news.

All the children used paper-based storyboards to facilitate their storytelling process. We identified three main patterns in their use: (1) drawing first, then telling (writing) stories; (2) telling first, then drawing; and (3) alternating between drawing and telling. More than half of the lower-grade children adopted the first strategy, while more than half of the higher-grade preferred the second one.

The children’s interviews further revealed three main roles of drawing in facilitating storytelling. **First**, more than half of the children said drawing helped visualizing abstract texts and enrich details, with P9 stating that “*Drawing can provide rich details*”, while others (P1, P4, P17, P3) mentioning drawing made stories more “*concrete*”, “*vivid*”, “*emotional*”, and “*easy to understand*”. **Second**, about one-fourth found that drawing provided references and resources for storytelling, acting as “*resources*” and “*hints*”. Interestingly, two children who first drew and then rigidly told or wrote stories based on their drawings mentioned that drawing “*provided references for storytelling texts*” (P10), that “*made storytelling more accurate*” (P2). **Third**, a few said drawing helped “*organize story structure*” (P6, P15) or “*relate the current story content to prior ones*” (P13).

However, children also acknowledged challenges with drawing, such as requiring “*skillsets*” (e.g., P7, P20), being “*time-consuming*” (e.g., P1, P18), “*difficult to change*” (e.g., P14, P19), being “*sometimes*

hard to represent texts” (e.g., P10, P25) or “*match with texts*” (e.g., P13, P26). We indeed observed that two children spent too much time drawing and coloring, preventing them from completing the storytelling tasks.

7.2.2 Storytelling Strategies with the AI System. In the AI condition, the top selected characters were a *scientist* (25.00%, $n = 10$ out of 40), followed by a *cat* (17.50%, $n = 7$), a *little boy* (17.50%, $n = 7$), and a *robot* (17.50%, $n = 7$). The top selected scenes were the *forest* (24.17%, $n = 29$ out of 120), *lab* (20.83%, $n = 25$), and *school* (13.33%, $n = 16$). Most children ($n = 15$) planned their emotional line as a common “positive-negative/neutral (turning point) -positive” emotional storyline (e.g., happiness-angry/surprise-happiness). Many ($n = 7$) started with a happy/negative emotion and ended up a neutral (surprise) emotion for an open-ended closure. A few ($n = 5$) chose to start with positive/neutral emotion but end with a certain negative emotion, while some ($n = 6$) started with a negative/neutral emotion but turned into a positive one. A few ($n = 5$) selected to maintain the same emotion (i.e., happiness) throughout the story. Many children indicated that they actively planned or partially planned their story structure in advance ($n = 22$) while the rest ($n = 16$) began planning upon seeing AI-generated keywords. Of the children, 87% ($n = 33$) indicated that they treated themselves as the character and used personal experiences and media sources such as novels or movies for storytelling references.

We observed three main patterns for children’s use of AI generated keywords. **First**, several children (e.g., P3, P11, P15, P29, P35) used the provided keywords without updating them to start, expand, or enrich their storytelling. This might be because they had few initial ideas, and the AI keywords offered good recourses to start. They were also willing to adjust the story direction based on the AI keywords. **Second**, most children (e.g., P1, P5, P8, P14, P16, P21) kept updating the keywords and strategically looked for specific ones they had in mind. However, they remained open to other options if they could not find the desired keywords for several rounds, or if they found something new and interesting. For example, P21 was looking for the word “*witch*” but noticed the “*forest elf*” keyword; he found it interesting to add a new character, so he actively adjusted his storyline. **Lastly**, a few children (e.g., P10, P13, P32) also kept updating the keywords, but were reluctant to change their ideas until the facilitator suggested it, indicating these children were more attached to their original ideas during storytelling.

Similar to the pilot testing conducted, when the children were initially asked to use the AI-generated keywords for story creation, most found it challenging. After the facilitator provided some guidance (e.g., “*think about how the character got the object and arrived at the scene.*”), many started to shift their strategy from changing keywords to purposefully thinking, which made them gradually become more confident with the process. The system provided four strongly associated keywords and four weakly associated ones, although the children did not know their categories. We found that most children preferred the strongly associated keywords. The post-interviews conducted with the children revealed their justifications for keyword selections. About one-fourth of the children looked for words with close relevance to the storylines and reasonable logic when integrated into the storylines, while around 10% preferred

words with novelty and inspiration. More than half sought words that were reasonable and simultaneously introduced new details or ideas. Most children expressed a great sense of accomplishment and satisfaction in using these nonsensical keywords to create coherent stories.

Regarding the AI-generated images, we noticed that all the children engaged extremely well in image generation. Many (e.g., P7, P9, P11, P13, P17) paid attention to details, and would ask for the regeneration of an image if it did not meet their expectations, such as missing important items, or having inappropriate size, position, or actions of the main character. We observed two ways that children strategically used AI images. **First**, most children used visuals to represent their story text like the traditional method; but they took a more iterative approach to gradually align text and images (**iterative visual representation**). Specifically, several children (e.g., P18, P33) added detailed descriptions for the visual parts that they were not satisfied with for regeneration. For example, P33 added details such as “heavy footsteps” and “a big shadow” for a few rounds, to center the main character dragon in the image. Some children even attempted to guess reasons and strategically adjusted the text to regenerate images. For example, P31 used the word “小花” (pronounced as /xiao3, hua1/) to represent “little flower”. When the AI image showed a calico cat instead, she realized the AI might interpret “小花” (short for “小花猫”, pronounced as /xiao3, hua1, mao1/) as a common cat name in China. She then changed the word to “花朵” (pronounced as /hua1, duo3/) for flowers and got the expected image. **Second**, a few children leveraged the images to enrich and adjust their storytelling (**story elaboration and inspiration**). Different than the traditional method where images mainly served as references, these children added unexpected story elements based on AI images. For instance, when P17 saw an orange cat generated by AI, he later described his character as a “cute orange cat” instead of just “cat”. P5 originally intended for an alien to be a villain who kidnapped a child robot. When he noticed that the alien in the AI image appeared adorable and kind, he decided to portray the alien as a kind-hearted character who helped the robot’s mother search for the child robot. However, AI images sometimes could restrict a child’s storytelling. We found that when a few children (P1, P14) could not generate the expected images after several attempts, they reduced the storytelling text to better match the image instead.

7.3 Children’s Preferences and Teachers’ Attitudes

When we asked the children about their favorite parts on the storyboard method, all of them nominated drawing, given its roles in visualization, inspiration, and referencing. However, more than half ($n = 20$ out of 38) felt they were not skilled at drawing, that drawing was time-consuming, and that they could not fully express their ideas. All the children ($n = 40$) after the AI session expressed a strong willingness to continue using the system in the future. We should carefully interpret this positive feedback due to potential novelty effects and children’s prior positive attitudes towards AI.

Regarding their favorite system functions, most children mentioned the AI-generated images ($n = 33$ out of 40), followed by AI-generated keywords ($n = 8$), character/scene selection ($n = 8$),

and the audio-text function ($n = 7$). As explained by the children “it was so convenient to quickly generate elegant images” (P30). Or “I liked the novel images generated by AI but was not satisfied with the quality of some images” (P38). Or even, “being able to freely choose the character was great. . . [I knew I] . . . should first think about the character, and it would make storytelling easier” (P18). There is also the view that “the regenerated keywords provided me inspirations...When connecting all the keywords, I could quickly make a story” (P22) and “the keywords provided inspiration and guidance for me to create the story” (P40). The results were consistent with our observations that most children initially found using these keywords challenging, but once they tried and succeeded, they became more confident. Interestingly, when we asked for suggestions for the system, most children mentioned the same three preferred functions. Eight children noted that AI-generated images differed from their expectations (e.g., smaller size of the character, lack of coherence), six suggested to add more personalized keywords, and three suggested to add more options for characters and settings.

The teacher interviews demonstrated strong interest in using the AI system to facilitate children’s creative storytelling. All three teachers believed that the system’s design, particularly the AI-generated images, would engage children and increase their motivation. Allowing children to select preferred characters and scenes and arrange the narrative structure in advance was also beneficial. The flexibility of selecting keywords, without restrictions on their relevance or actual use, enabled easy integration of the system into real teaching contexts, as teachers could design various learning tasks according to children’s storytelling levels. However, the teachers noted that inconsistent generation of the main story character by AI might confuse children occasionally. They suggested it would also be beneficial if the system allowed more control over image generation.

8 Discussion

This research aims to understand how to design effective AI-empowered storytelling systems for elementary children (RQ). To achieve this, we derived design requirements from formative research and validated them through the iterative designs and evaluations of a research prototype. In particular, we compared children’s creative storytelling quality (Rq1), strategies (Rq2), and preferences, as well as teachers’ attitudes of the system (Rq3) compared to a traditional paper-based storyboard method. We will now synthesize prior theoretical and empirical work with our findings, to discuss the advantages and considerations of using generative AI in supporting children’s creative storytelling.

8.1 Promoting High-Quality Creative Storytelling: Balancing Structure Planning and Emergent Details

A good story typically has a mature and potentially complex narrative structure planned in advance [32]. By guiding children to plan the global story structure first [2], then using AI to fill gaps with serendipitous details, can enhance storytelling quality, particularly creativity and richness (Rq1). The system allows children to plan their stories by selecting characters, arranging scenes, and designing emotional trajectories. During the process, AI-generated

keywords and images based on prior storylines and new inputs can create an “emergence and constraint” through algorithmic uncertainty, which serves as a central mechanism for creativity which is emphasized by [17] adding unexpected details or new directions for storytelling.

Before storytelling, a few children in the storyboard condition planned their structure, while in the AI scenario, all did so with the AI system (with the system’s core design feature). During storytelling, children using storyboards did not have external mechanisms of emergence for producing alternatives but heavily relied on their creativity and literacy abilities. While collaborating with the system, the AI keywords and images created constraints (e.g., selecting one keyword to integrate into the storyline) and provided rich cultural and historical resources for children to develop and elaborate their ideas. Children ended up producing higher-quality creative storytelling with the AI system, particularly excelling in story creativity and richness. The richness reflected not just word count but also detailed content elaboration. This was consistent with our observations that children paid attention to AI-generated image details, which was hard to achieve with storyboards. Such a design mechanism may shed light on other AI-based educational technologies for problem-solving tasks (like storytelling). Enabling children to create an overall plan and use it to guide child-AI collaboration may help to reduce the side effects of uncertainty and lack of transparency in AI-generated content.

Interestingly, although fewer children planned their story structure with storyboards, they scored better in terms of coherence and narrative structure. We interpret this as follows: (1) Many children wrote complete sentences on storyboards for reference aiding coherence. By considering this point, we asked the experts to evaluate coherence based on content rather than conjunction vocabularies, though writing itself may help thinking. (2) Paper-based storyboards with physical affordances allowed easy review of notes and sketches, aiding in establishing a holistic view. However, children in the AI condition were only given a chance to check their previous storytelling at the end. (3) The AI keywords and images created a strong emergent power that may drive the planned storyline in a new direction. Therefore, designers should: (a) balance structure planning with emergent creativity by aligning AI with children’s plans, ensuring their shared goals prior storytelling; and (b) encourage frequent content review by adding system navigation and reminders or making storytelling elements tangible as suggested [20], so that children can easily review and continue refining their stories, even if off the system.

8.2 Leveraging Generative AI for Iterative Creative Storytelling: Considering Blending Visual and Text Prompts

Unlike previous generative AI storytelling systems that focus on either visual storytelling [2, 16, 36] or generating purposeful questions for parents [37], we leveraged generative AI to provide both fragmentary verbal (words rather than the entire sentences/paragraphs) and nonverbal (visual images) prompts to promote children’s diverse creative storytelling strategies (Rq2). We observed that AI generated words provided most children with resources to initiate, ideate, enrich, or expand their stories while offering

some children challenging tasks for them to reflect, rethink more possibilities with nonsensical words, and potentially change the original direction, both of which can benefit children’s creativity. We also noticed that generated images engaged children, helped them to visualize abstract content, and sometimes inspired them to change their storyline with serendipitous visual elements. This suggests the effectiveness of using both AI-generated texts and images, especially offering fragmentary AI cues, in supporting children’s creative storytelling.

Additionally, we noticed that children in both conditions attempted to closely relate visuals and texts during storytelling. Whenever starting with visuals or texts, they explored ways to best merge the two. For example, to generate the desired AI visuals, children might have iteratively revised their narrations to be more specific about details. These behaviors were noted in and suggested to be beneficial for enriching creative details [36]. When using storyboards, most children began by drawing to visualize stories, inspire ideas, and organize the structures. However, the limitations of hand drawing were also apparent (e.g., requiring skills, difficulty in representing abstract and complex content, and not being easily changeable), which restricted children from further adjusting their ideas dynamically according to the situations. We even noticed several children rigidly using drawing materials as visual references for storytelling (i.e., rigidly describing the story based on the visual elements), which was not conducive to producing alternatives and was not good for creativity.

In contrast, in the AI system scenario, the AI’s capability to quickly generate strongly/weakly associated keywords and vague/unexpected visual images sometimes became an advantage in stimulating creativity. Children could start with simple sentences and generate images for quick checks. The AI’s effectiveness in producing a series of images also encouraged children’s quick reflection and relation to their storytelling texts, allowing for timely adjustments and image regeneration. By blending AI visual and text prompts, the system can facilitate a more iterative and cyclic process for thinking, generating, and refining storytelling, rather than the linear text-to-visual or visual-to-text processes.

8.3 Enabling Individual Satisfaction: Increasing Contextual Flexibility and Individual Agency in Child-AI Interaction

The system was designed to facilitate elementary children’s creative storytelling across grades supported by adults such as their teachers or parents (Rq3). To accommodate children with varying literacy skills, it is important to maintain design flexibility for contextual use. Taking the system as an example, teachers can adjust task difficulty by changing the setup of the character, settings, and emotions (e.g., a story with a “happy-sad-happy” emotional trajectory may be easier compared to an “angry-sad-surprise” for most children). Encouraging the use of strongly or weakly associated words could be another strategy, although we noticed that children in our study tended to select more strongly associated keywords. Although we have used prompts to ensure the suitability of the keywords generated by ChatGPT for children, involving adult experts in the contextual use of AI-based educational technologies is necessary

for ethical considerations, due to the opaque and uncontrollable nature of AI.

Children expressed a strong desire for more personalized keywords, more “accurate” images matching their expectations, and more options for characters and setting, all indicating a desire for greater control and agency. The importance of ownership in child-AI collaboration has been discussed in several prior works [2, 16]. However, we noticed there may be conflicts between learning and engagement regarding the distribution of ownership, and this distribution of ownership should be tailored to specific tasks. For example, we observed children’s interest in exploring various uses of keywords and their sense of accomplishment after successfully integrating given words into their storylines. This suggests effectiveness, by assigning AI the agency to provide keyword constraints to boost children’s creativity. Compared to this, although many children preferred the AI image (re)generation functions (motivation), only a few actively used it to inspire or enrich their storytelling, and most expressed dissatisfaction with visual representations due to inconsistency and inaccuracy. Particularly, several children preferred to rigidly visualize their storytelling, showing the limitations of relying heavily on visuals for storytelling. Additionally, despite the system enabling children to regenerate images, they seemed to prefer fine-tuning details of the given ones rather than completely changing them. We noticed several children with much AI experience attempted to guess the underlying mechanisms and used strategies to fine-tune their images, indicating a request for more agency from AI in image control. Therefore, we suggest designers consider dynamically adjusting the balance of agency between children and AI in their collaboration process, considering both learning effectiveness and the use experience for various sub-tasks. For example, in storytelling, designers could allow children to define or select image keywords and artistic styles. This agency may also enhance children’s strategic use of visuals in storytelling, which was less observed in our study.

9 Limitation

We acknowledge several limitations of the study. First, due to a limited number of facilitators, each child in the traditional group was not assigned an individual facilitator. This led some children to write complete sentences as storytelling references. To address this, we evaluated coherence based on content rather than conjunction words. Second, to compare storytelling quality between conditions, we provided a maximum 30 minutes for each child, which seemed insufficient, especially for higher-grade children. Third, we did not conduct a longitudinal study to assess development. Particularly, we could not determine if higher scores were due to AI involvement or the enhanced abilities from the children. While we observed some evidence of transferable learning when many children expressed interest in applying the keyword method to future storytelling, longitudinal evaluations remain necessary. Lastly, we relied on self-evaluations for children’s prior storytelling and creativity levels and did not examine the impacts of children’s age differences, personal characteristics, prior abilities, and diverse performances on storytelling quality. These aspects are beyond the current study’s scope and will be addressed in future research.

10 Conclusion

We present the design and evaluation of an AI-empowered creative storytelling system for elementary children. We conducted formative research with teachers and children and derived five design implications for developing AI-empowered storytelling tools for children. Based on the findings, we developed an interactive system named *StoryPrompt* that enables elementary children to co-create stories and comics with generative AI, fostering their literacy and creativity through storytelling practices. Pilot testing with eight children and three HCI experts demonstrated good system usability and satisfactory learning experiences. The research further explored how 40 children used the AI system and the traditional storyboard method for creative storytelling. We also interviewed three teachers to gather their feedback on the two methods. Quantitative results showed significantly higher storytelling quality with the AI system, particularly in creativity and richness. Observations and interviews revealed more purposeful storytelling planning and strategic use of AI-generated words and images, allowing children to efficiently explore storytelling alternatives by ideating, expanding, or modifying content. Children in both conditions preferred visual images but noted their limitations, highlighting the inefficiency of relying solely on visual storytelling. Teacher interviews highlighted the AI system’s potential to motivate learning and its flexibility for classroom integration.

To achieve high-quality storytelling, we suggest (1) leveraging both AI-generated text and visuals to facilitate creative storytelling, promoting an iterative and cyclic thinking process rather than a linear one, and (2) carefully balancing the power between the overall story planning and the emergent details offered by both generative AI. To increase satisfaction, it is important to maintain the contextual flexibility of the system’s design, supporting adaptable use by children and teachers in school learning contexts. Designers should also balance the agency of children and AI to ensure effective and satisfactory learning experiences.

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APPENDICES

A. 1 Interview Protocol with Teachers

- Do elementary children receive creative storytelling training at schools? If yes, when do they start? What are the learning goals (e.g., language, creativity, problem-solving)? Do the learning goals vary for children in different grades, and how?
- Are there different types of storytelling activities? What are the processes of these storytelling activities like?
- What are the effective or frequently used training methods?
- [Do children often encounter any problems? What are these problems? What scaffolding can be offered to children?
- What digital devices/equipment does the school have? What are the digital literacy levels of teachers and children? Can children in various grades use computers independently?

A. 2 Focus Group Protocol with Children

1. What types of books do you like? What are some books that you have recently read? Could you please provide any examples?
2. Do you engage in storytelling activities with your parents after school? If yes, when and how?
3. If you were asked to create a story, what kinds of characters would you like to choose as the main character for your story? Could you give us 3-5 examples?
4. What kinds of scenes would you like your story to take place in? Could you give us 3-5 examples?

5. Have you ever used voice interaction in any digital devices (e.g., smartwatch)? Have you ever played computer games/software? Can you use a computer mouse and keyboard?

B. 1 Semi-Structured Observations and Interviews with Children
Semi-structured observational sheets

1. How do children start their stories, first with visuals or first with verbal (text)?
2. Do children have any challenges during storytelling? How do they address these challenges?
3. Do the learning goals vary for children in different grades, and how?
4. How do children interact with storyboards and the AI system?
5. Are there any other interesting behaviors?

Several questions in semi-structured interviews are slightly adjusted according to the context of each condition.

1. How do you start your story? Do you plan the story before starting? Do you have any resources as references?
2. Do you have any challenges during storytelling? How do you address them?
3. What types of strategies do you use during storytelling?
4. How effective do you think the method is in supporting your storytelling?
5. How do you like this method? Which part of the method do you like best?
6. What other feedback would you like to provide?

B. 2 Semi-Structured Interviews with Teachers

1. What do you think about the two methods in supporting storytelling? What are the pros and cons?
2. Do you have any other feedback regarding the AI system?